

# OUT-OF-DISTRIBUTION AND ANOMALY DETECTION

IA Frameworks

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# LEARNING OUTCOMES

- Master the fundamental OOD / AD concepts
- Understand the main algorithm families, their advantages and weaknesses
- Familiarize with the most common evaluation metrics

# WHAT IS AN ANOMALY?

## Anomaly

"An observation that deviates considerably from some concept of normality"

# WHY CARE ABOUT ANOMALY DETECTION?

- Intrusion detection (cybersecurity)
- Fraud detection (finance)
- Industrial fault / damage detection
- Medical diagnosis
- **[NEW!]** Out-of-Distribution detection for ML and DL models

# MAIN CHALLENGES

- Ill-defined problem
- Absence of abnormal data
- Deep Learning use-cases:  
data is very high-dimensional

# WHAT IS AN ANOMALY?

"An observation that deviates considerably from some concept of normality"

A typical mathematical formalization:

$\mathcal{X} \subset \mathbb{R}^D$  data space

$\mathbb{P}^+$  ground-truth law of normal data on  $\mathcal{X}$  with p.d.f.  $p^+$

$A_z = \{x \in \mathcal{X} \mid p^+(x) \leq z\} \leftarrow \text{set of anomalies}$   


CAREFUL!

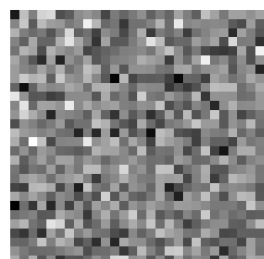
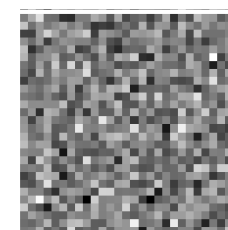
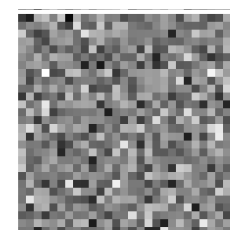
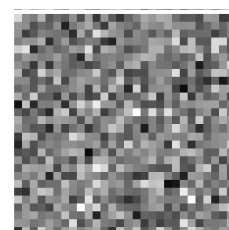
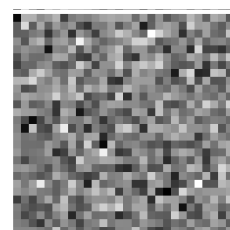
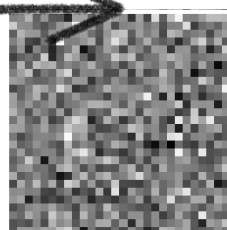
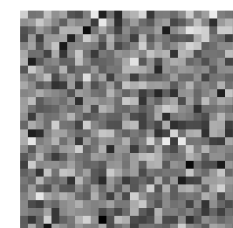
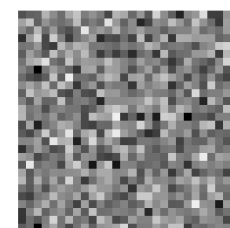
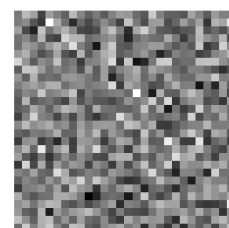
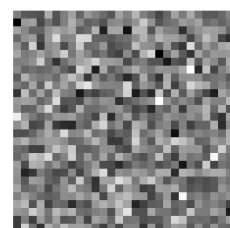
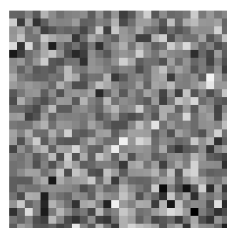
What is the concept of normality used in this formalization ?

CAREFUL!

What is the concept of normality used in this formalization?

10 NORMAL

images



Question: which of these 2 images is normal?

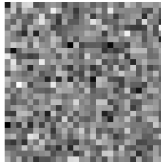
CAREFUL!

Yet: pixels are i.i.d.

$\mathcal{N}(0,1)$  random variables :

$\Rightarrow$  Likelihood Ratio

$$\frac{f(\text{smooth})}{f(\text{noisy})} = \prod_{ij} \frac{1}{e^{-\frac{x_{ij}^2}{2}}}$$

value of pixel  $ij$   
in 

# TYPES OF ANOMALIES

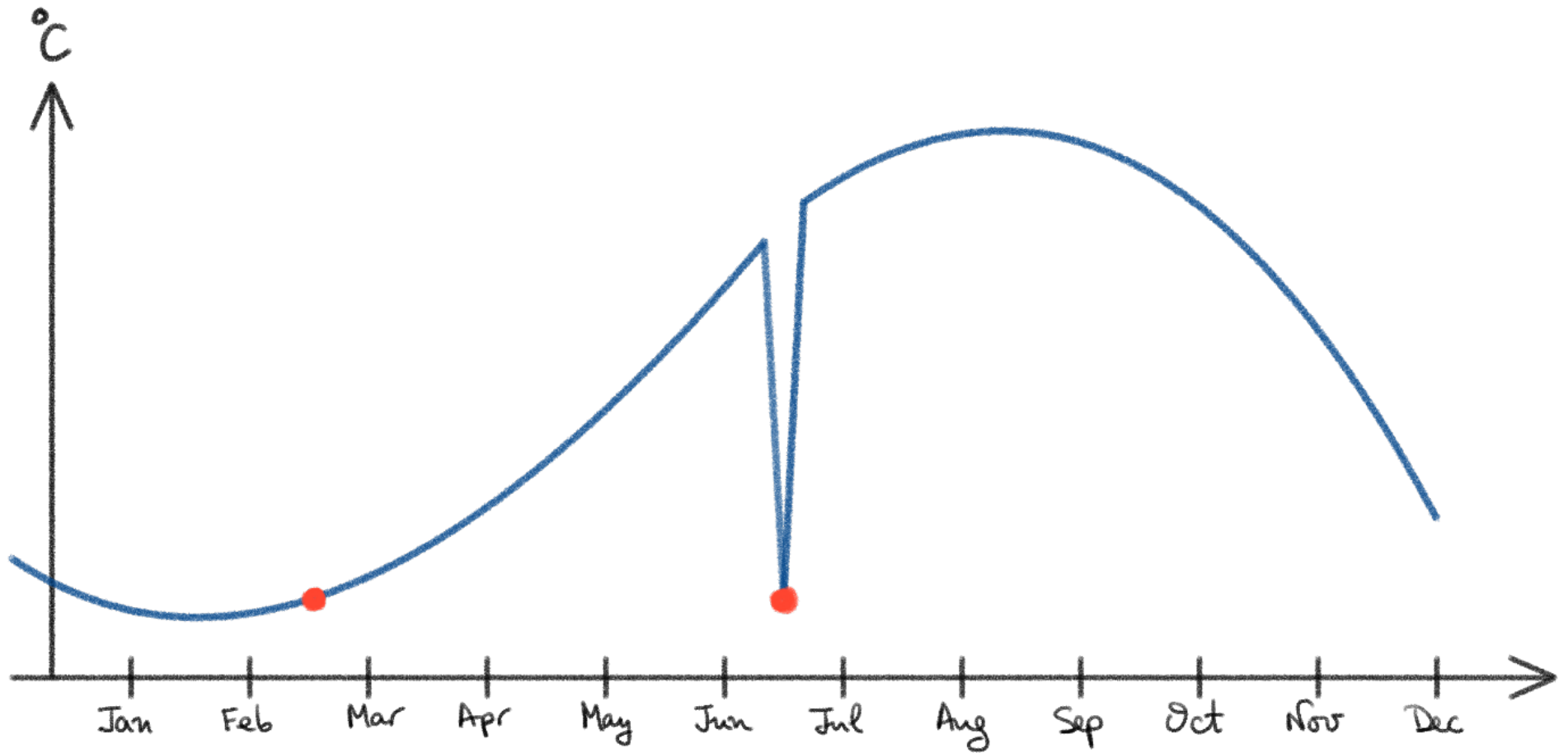
Traditionally:

→ Point anomalies

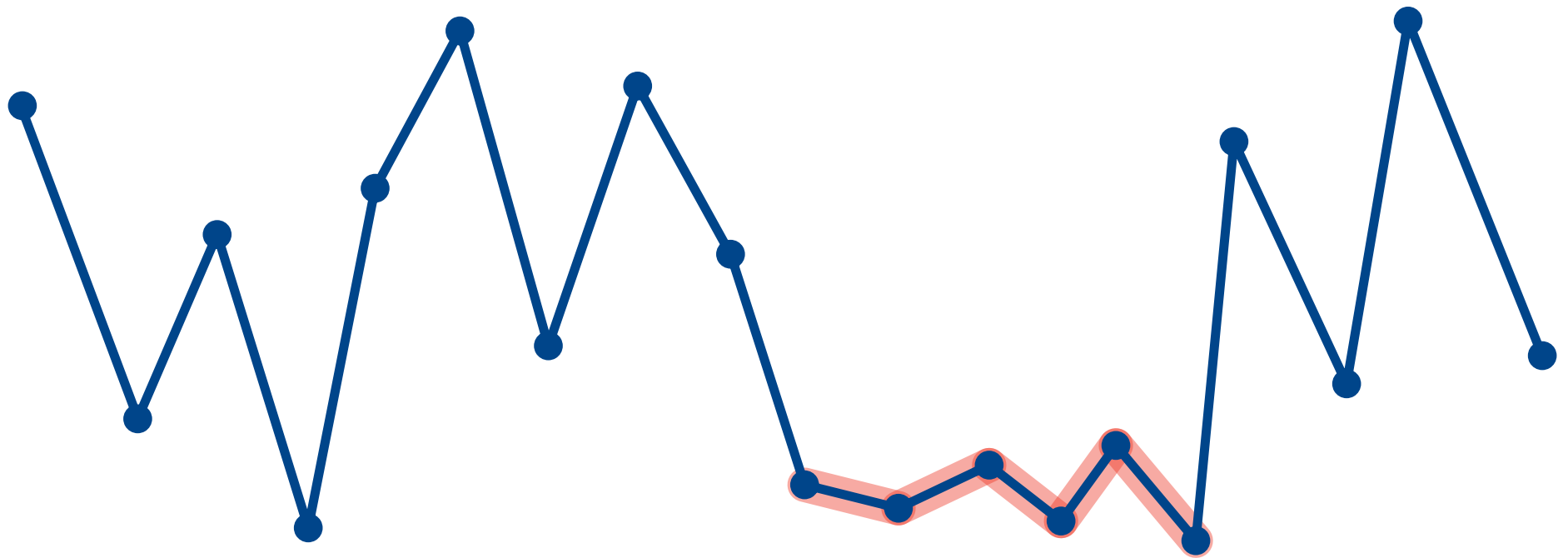
→ Conditional / Contextual anomalies

→ Group / Collective anomalies

# CONTEXTUAL ANOMALIES



# COLLECTIVE ANOMALIES



# TYPES OF ANOMALIES

Since Deep Learning:

→ Low-level sensory anomalies

→ High-level semantic anomalies

# TYPES OF ANOMALIES

Since Deep Learning:

- Low-level sensory anomalies
- High-level semantic anomalies

# LOW LEVEL, SENSORY



The book is  
interesting and  
it conveys a  
meaningful message.

# HIGH LEVEL, SEMANTIC



The girl was  
playing with a  
colorless green  
liquid.

# TYPES OF ANOMALIES

Anomaly not drawn from  $\mathbb{P}^+$

Outlier drawn from  $\mathbb{P}^+$  w low proba

Novelty from an evolving  $\mathbb{P}^+$

# ANOMALY DETECTION STRATEGIES

→ Model-agnostic

Given  $X_1, \dots, X_n \sim \mathbb{P}^+$ ,  $X_{\text{test}} \in \mathcal{A}_Z$ ?

→ Model-based

Given  $(X_1, Y_1), \dots, (X_n, Y_n) \sim \mathbb{P}^+ \times \mathbb{P}_Y$

and  $h: \mathcal{X} \rightarrow \mathcal{Y}$ ,  $X_{\text{test}} \in \mathcal{A}_Z$ ?

# ANOMALY DETECTION ALGORITHMS

## AD algorithm

$s: X \rightarrow \mathbb{R}$  and  $\tau \in \mathbb{R}$

such that

  $s$  is labeled "score" and  $\tau$  is labeled "threshold" in purple text with arrows pointing to them.

$\left\{ \begin{array}{l} \text{if } s(x) \leq \tau \Rightarrow x \text{ is declared "normal"} \\ \text{if } s(x) > \tau \Rightarrow x \text{ is declared "anomalous"} \end{array} \right.$

# ANOMALY DETECTION ALGORITHMS

→ One-class classifier

→ Probability density estimation

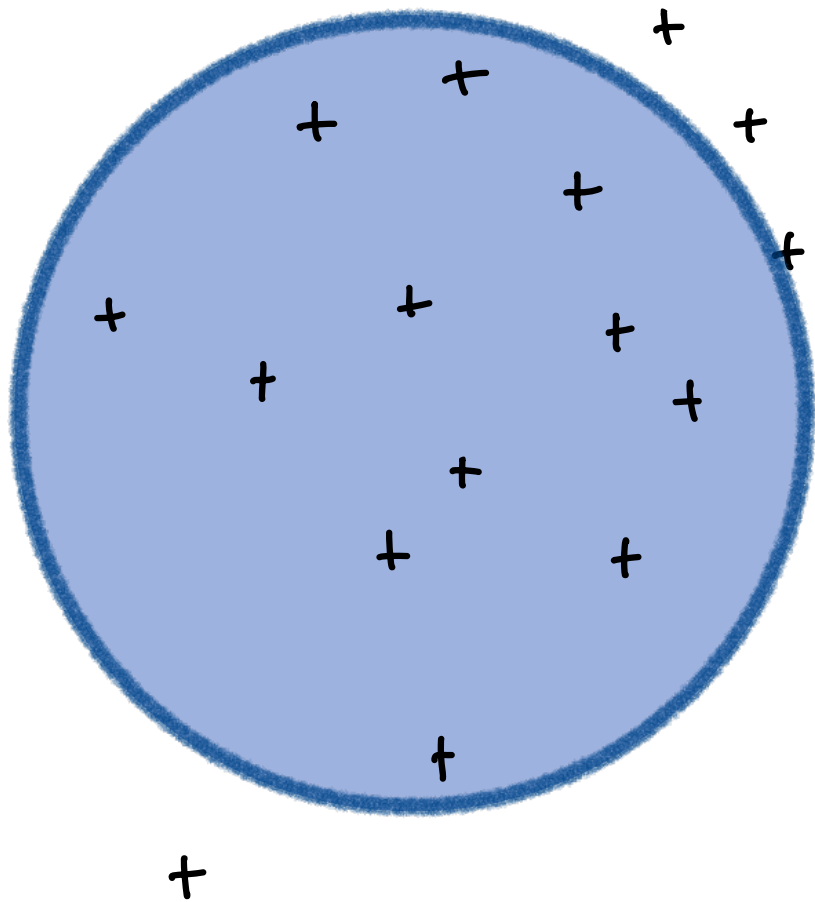
→ Reconstruction

→ Distance

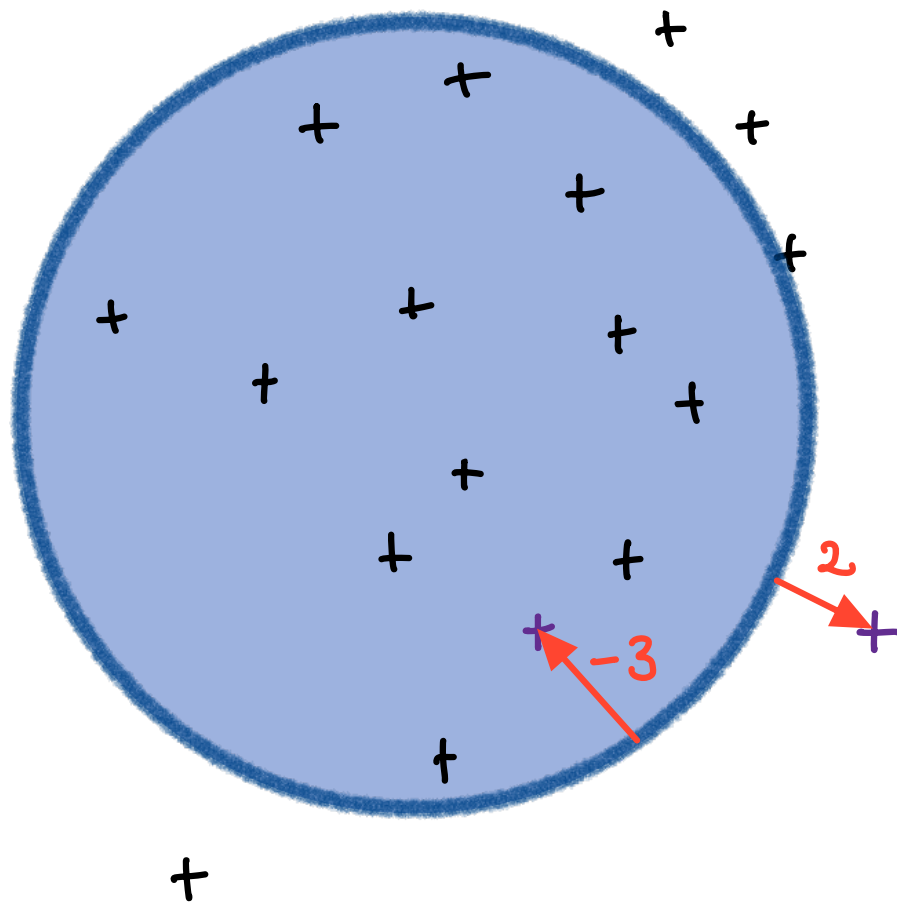
+ predictor  $h: X \rightarrow Y$

(if available)

# ONE-CLASS SVM



# ONE-CLASS SVM



$S(x)$  = signed distance  
from  $x$  to  $\bigcirc$

$$z = 0$$

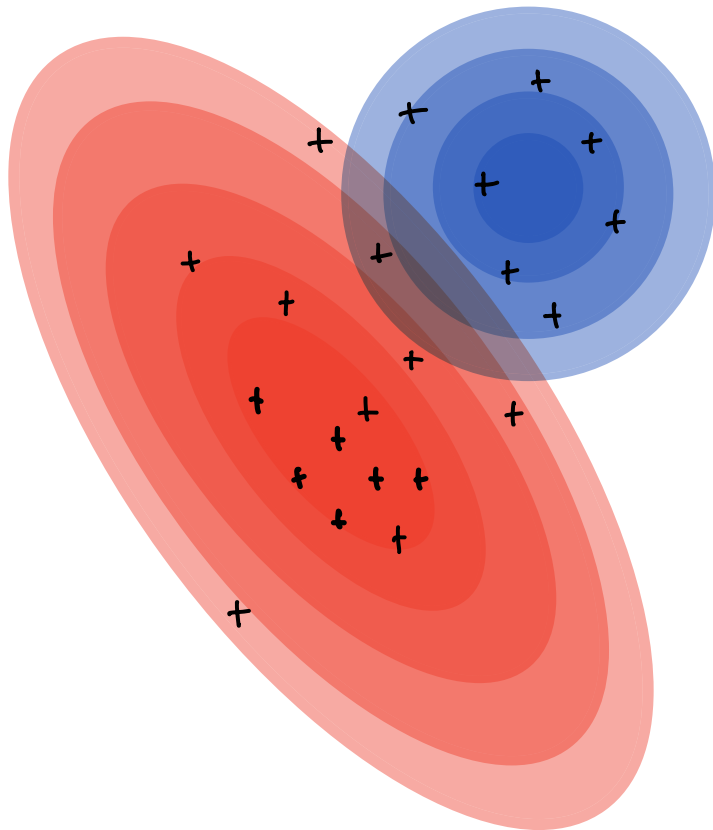
$$\Rightarrow \begin{cases} x \in \bigcirc \Rightarrow \text{"normal"} \\ x \notin \bigcirc \Rightarrow \text{"anomaly"} \end{cases}$$

# PROBABILITY DENSITY ESTIMATION

e.g. with a Gaussian Mixture Model (GMM)

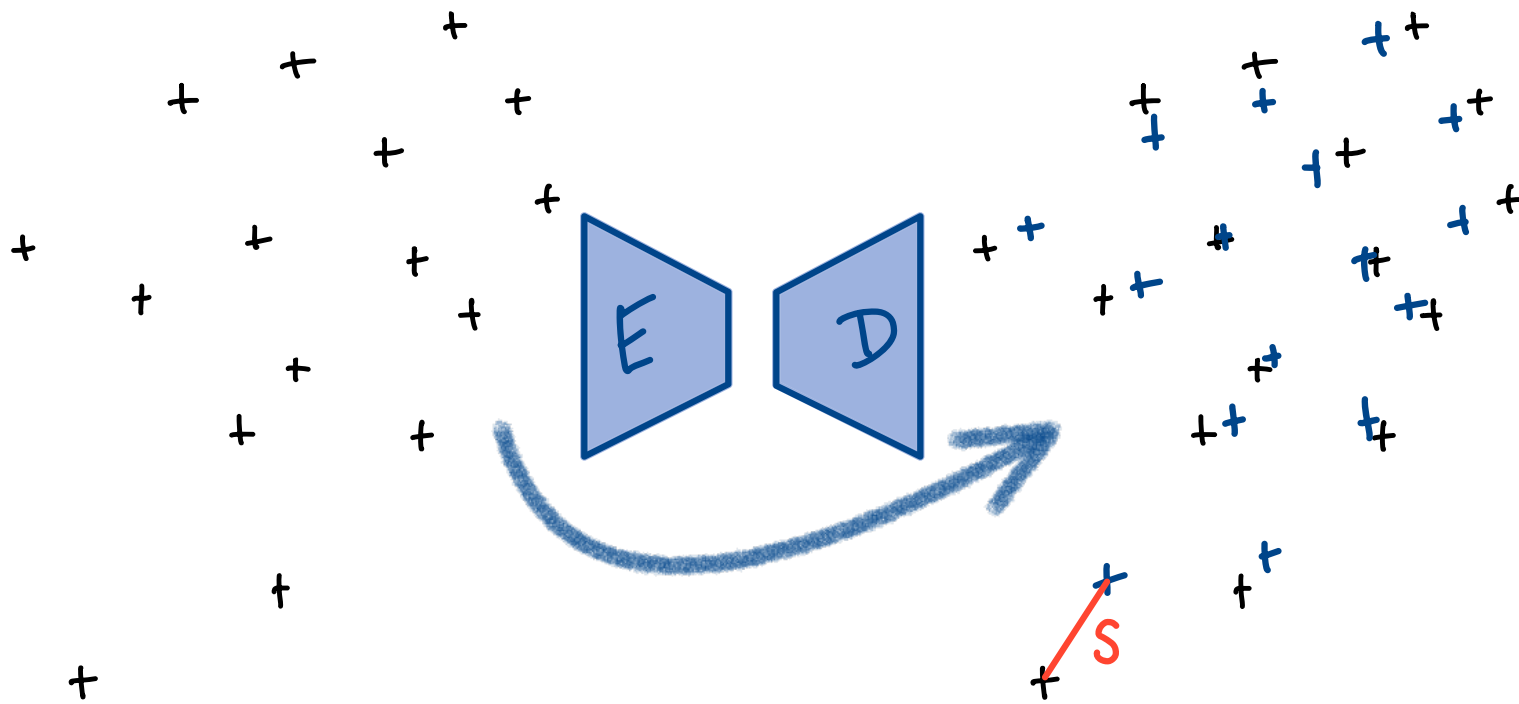
i.e.  $\mathbb{P}^+ \simeq \sum_{i=1}^K w_i \mathcal{N}(\mu_i, \Sigma_i)$

GMM fitted on  
 $x_1, \dots, x_n$



$S(x) = -\text{log-likelihood}$   
of  $x$  under the GMM

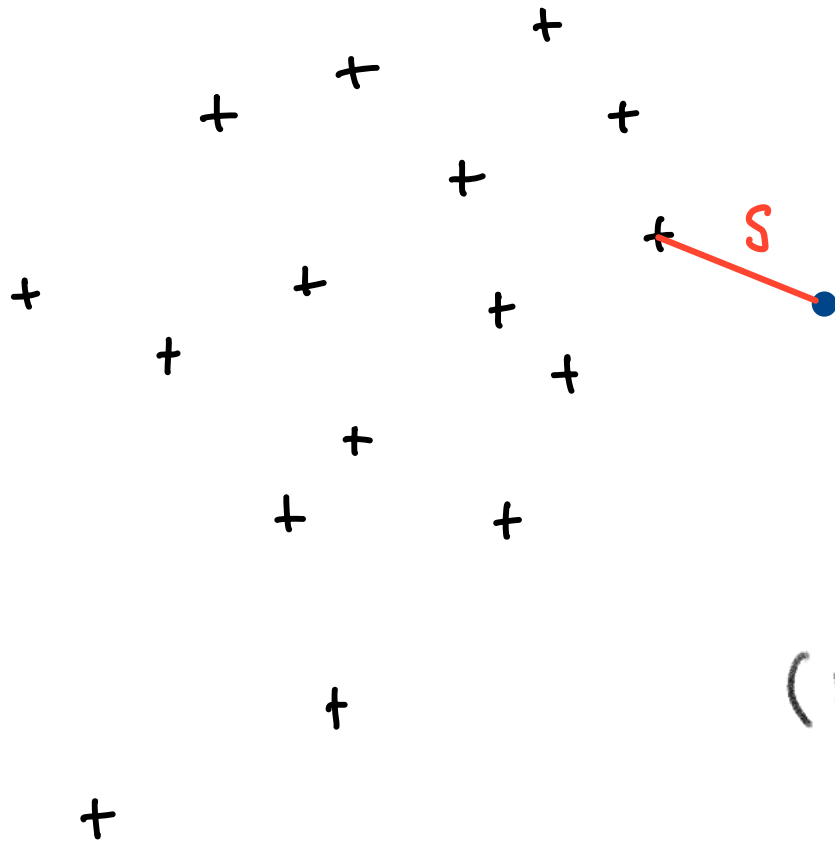
# RECONSTRUCTION W. AUTOENCODERS



$$S(x) = d(x, \mathcal{D}(\mathcal{E}(x)))$$

# DISTANCE W. K-NEAREST NEIGHBORS

$$+ \sim \mathbb{P}^+$$



$S(x)$  = distance to  
(K-th) nearest neighbor

# Quiz

|      | sensitive to outliers? | scales to large dim? | easy to pick? |
|------|------------------------|----------------------|---------------|
| SVM  |                        |                      |               |
| GMM  |                        |                      |               |
| AE   |                        |                      |               |
| K-NN |                        |                      |               |

# EVALUATION METRICS

Normal = -

Anomaly = +

|              |   | Predicted      |                |
|--------------|---|----------------|----------------|
|              |   | -              | +              |
| Ground Truth | - | True Negative  | False Positive |
|              | + | False Negative | True Positive  |

Confusion Matrix

# EVALUATION METRICS

For a fixed threshold  $z$ :

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

|              |   | Predicted      |                |
|--------------|---|----------------|----------------|
|              |   | —              | +              |
| Ground Truth | — | True Negative  | False Positive |
|              | + | False Negative | True Positive  |

$$\text{Recall / True Positive Rate} = \frac{TP}{TP + FN}$$

# EVALUATION METRICS

For a fixed threshold  $z$ :

$$F_1 \text{ score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{False Positive Rate} = \frac{FP}{TN + FP}$$

## Quiz

1. Is accuracy an adequate metric for AD? Why?
2. Give an example of a confusion matrix where precision and recall are both high.
3. Express the  $F_1$  score in terms of the elements of the confusion matrix
4. The  $F_\beta$  score is 
$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \beta^2 \text{Recall}}$$

What term do we give more importance to

(i) if  $\beta < 1$ ?      (ii) if  $\beta > 1$ ?

# SOLUTIONS

$$F_{\beta} = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \beta^2 \text{Recall}}$$

# EVALUATION METRICS

Threshold independent: AUC-ROC / AUROC

ROC = "Receiver Operating Characteristic"

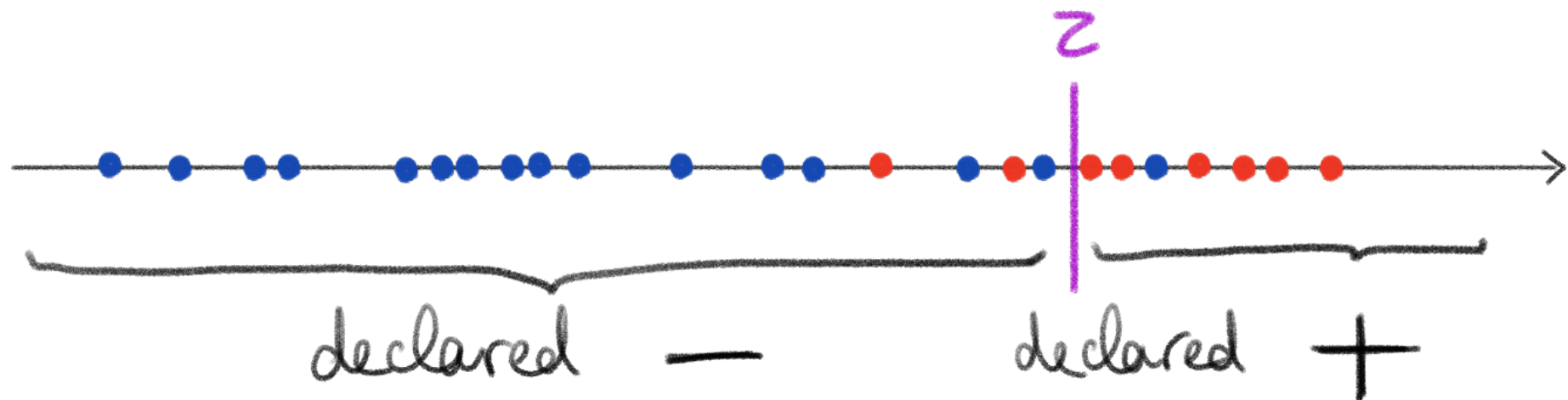
Given:  $s: \mathcal{X} \longrightarrow \mathbb{R}$

● = normal (ground-truth)

● = anomaly (ground-truth)



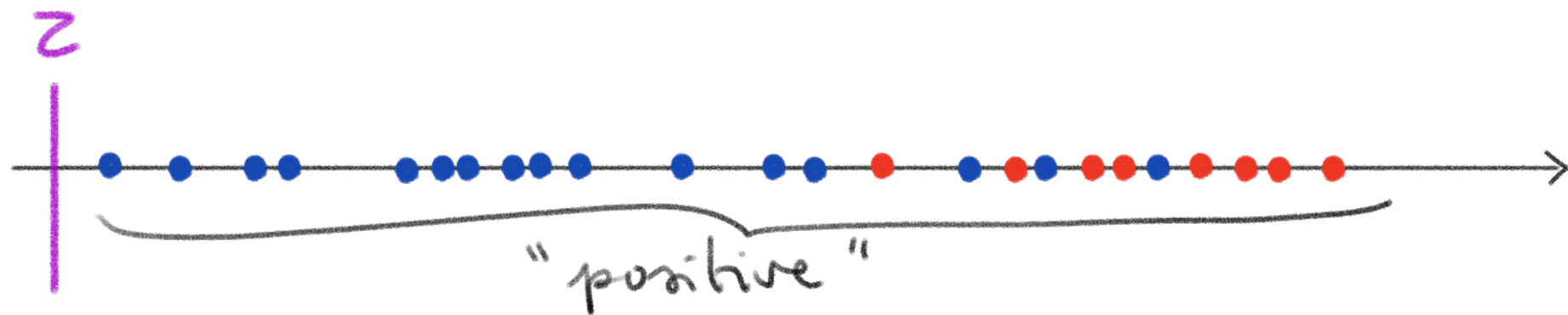
# EVALUATION METRICS



$$TPR(z) = \frac{\text{\# of red dots and +}}{\text{\# of red dots}}$$

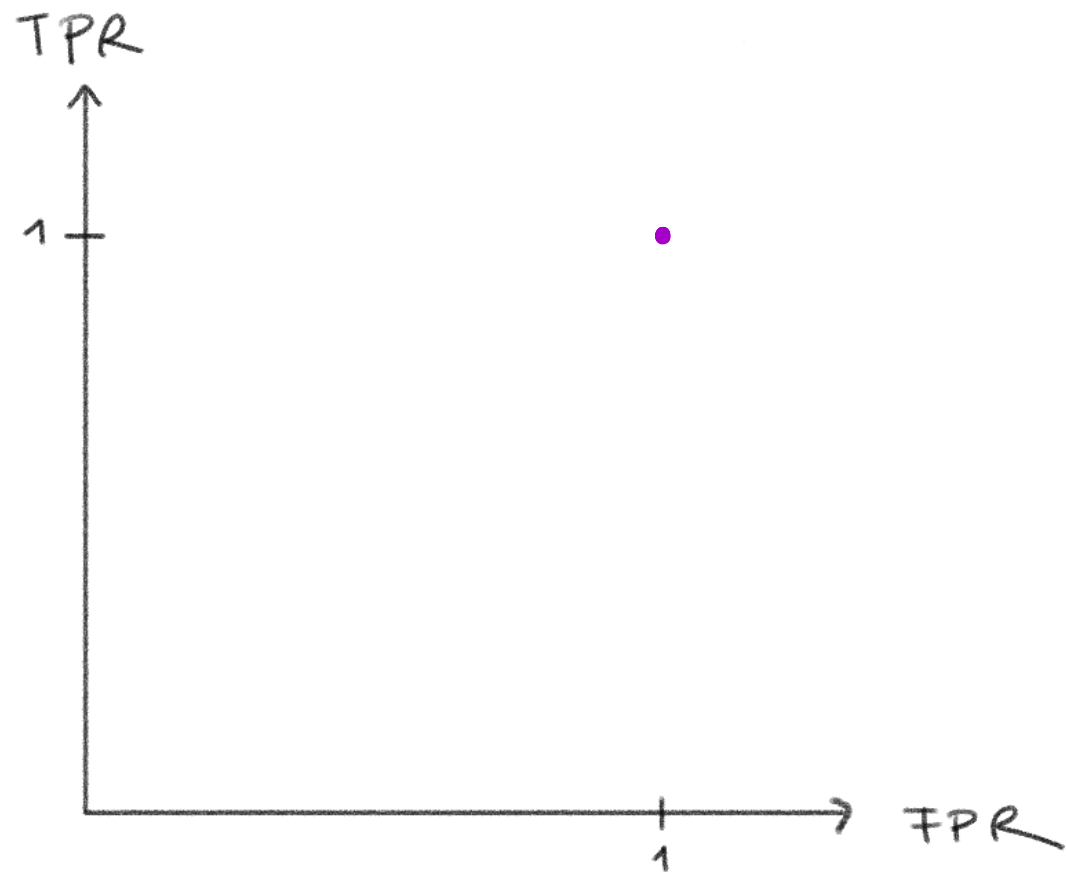
$$FPR(z) = \frac{\text{\# of blue dots and -}}{\text{\# of blue dots}}$$

# EVALUATION METRICS

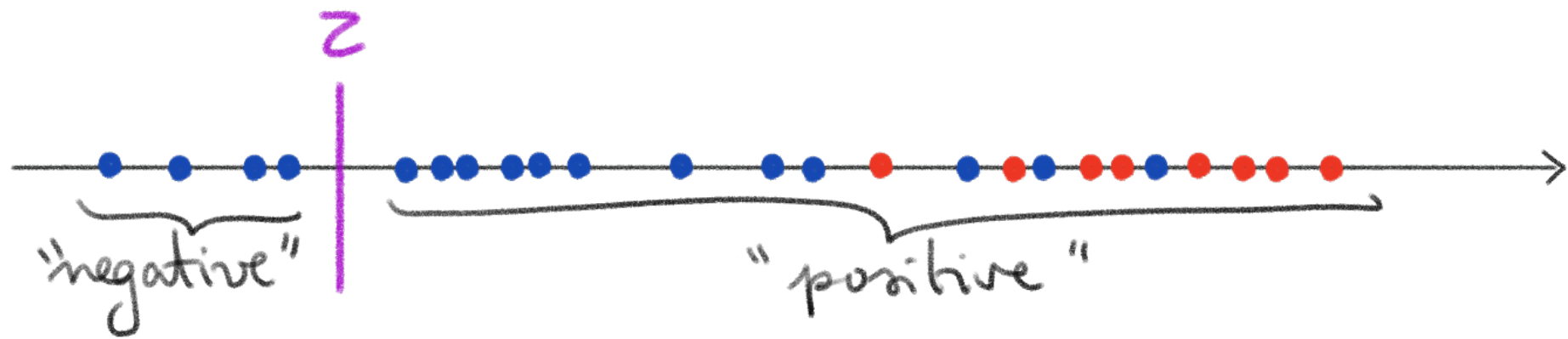


$$FPR(z) = 1$$

$$TPR(z) = 1$$

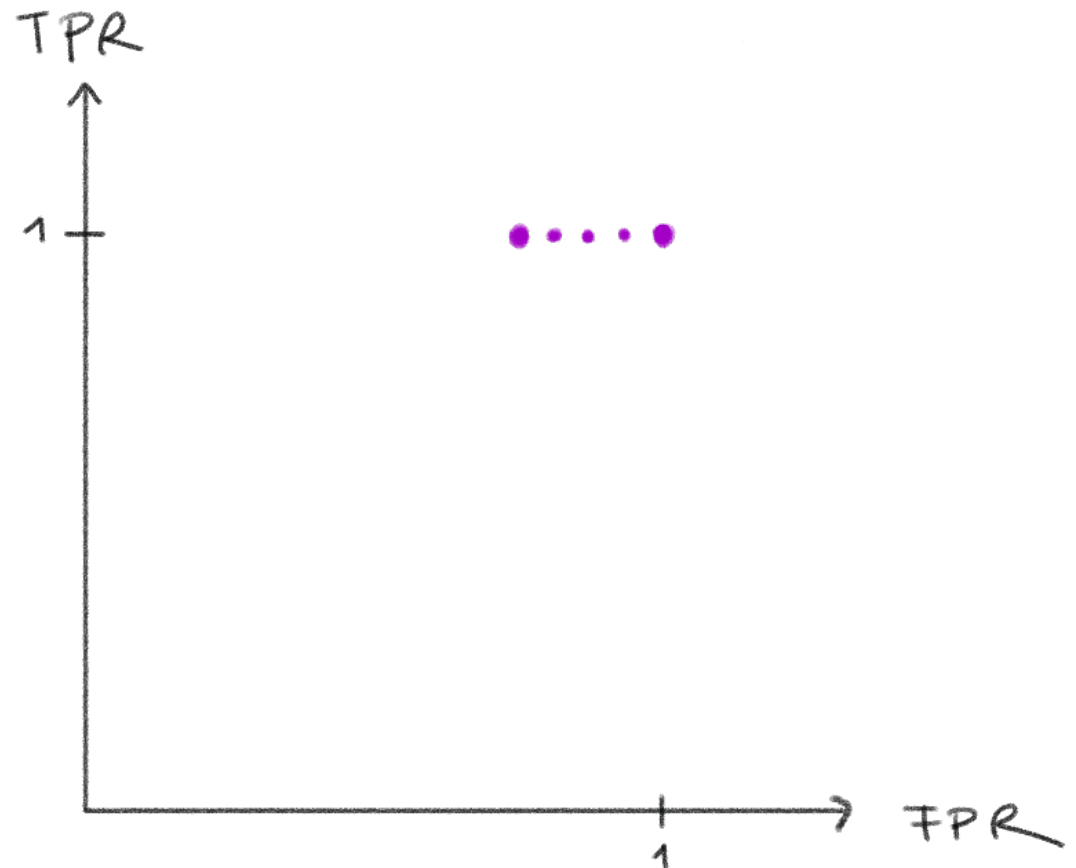


# EVALUATION METRICS

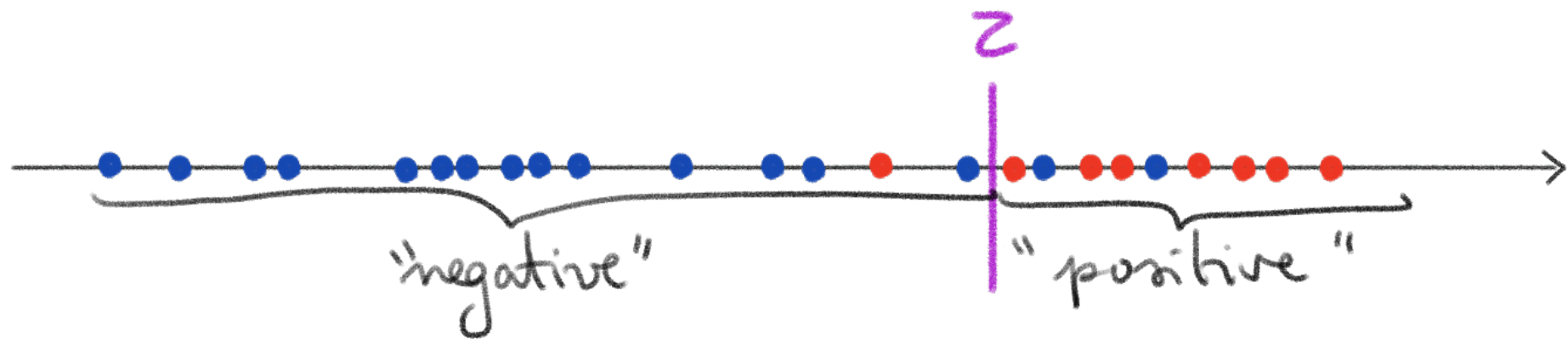


$$FPR(z) = \frac{12}{16} = \frac{3}{4}$$

$$TPR(z) = 1$$

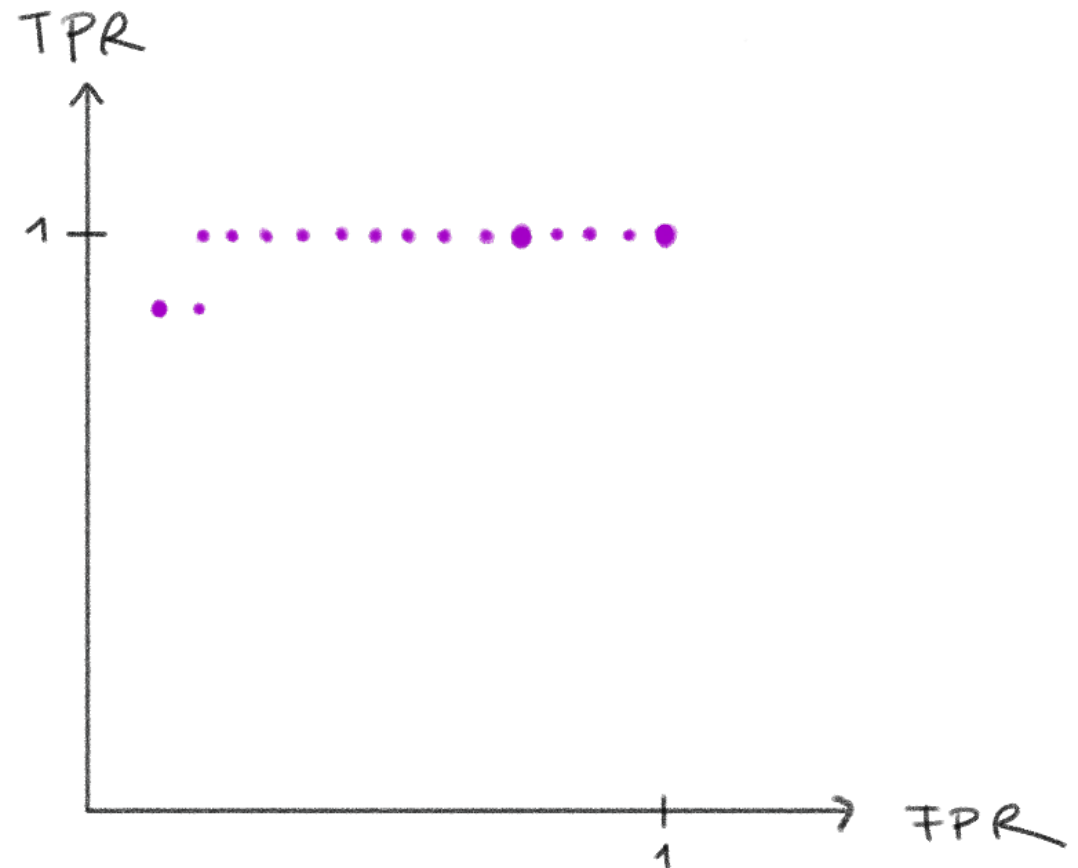


# EVALUATION METRICS

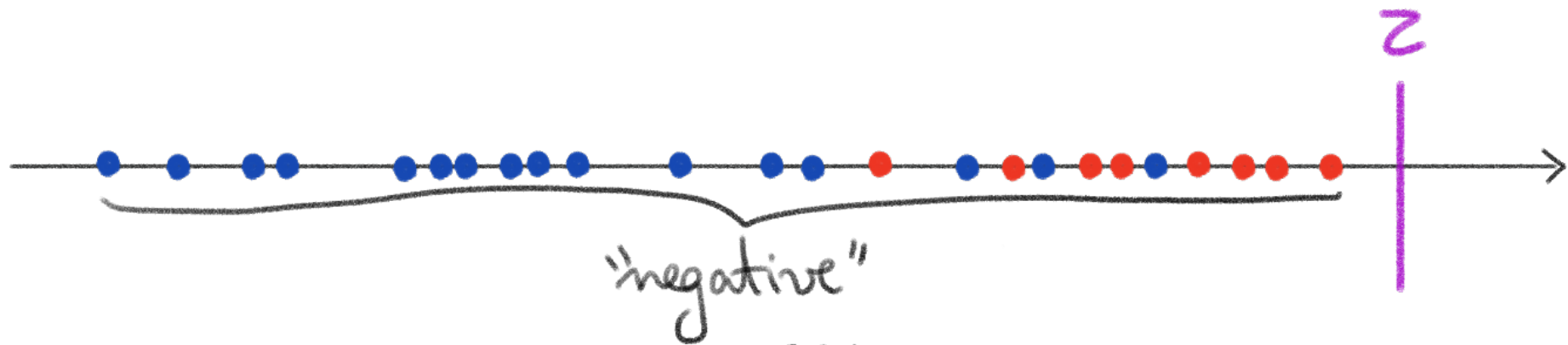


$$FPR(z) = \frac{1}{8}$$

$$TPR(z) = \frac{7}{8}$$

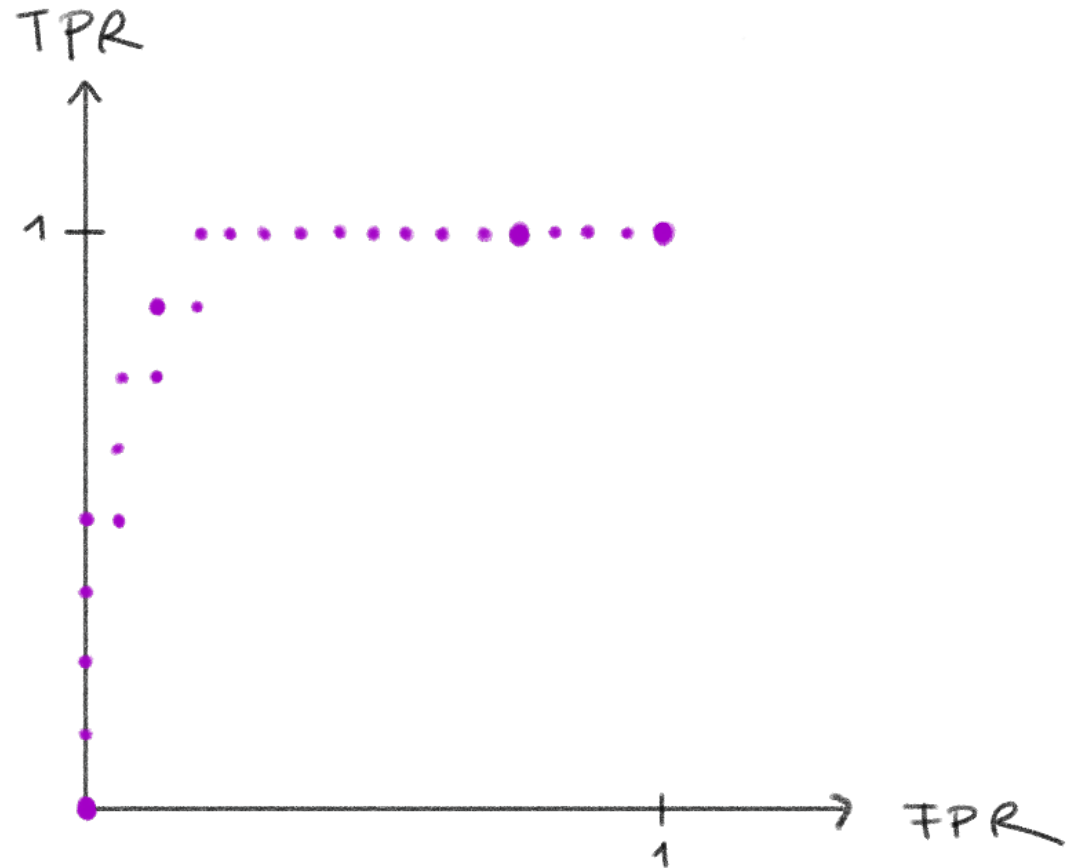


# EVALUATION METRICS

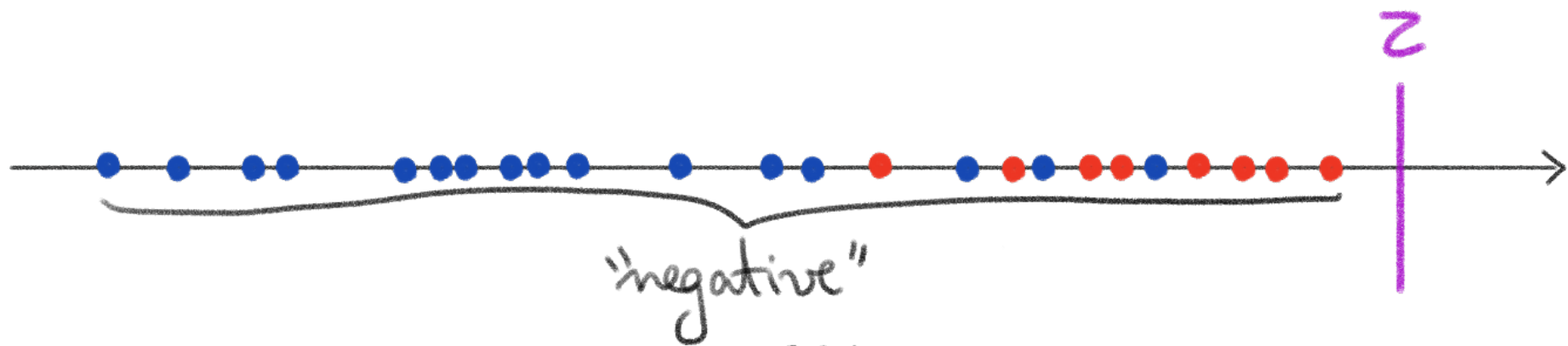


$$FPR(z) = 0$$

$$TPR(z) = 0$$

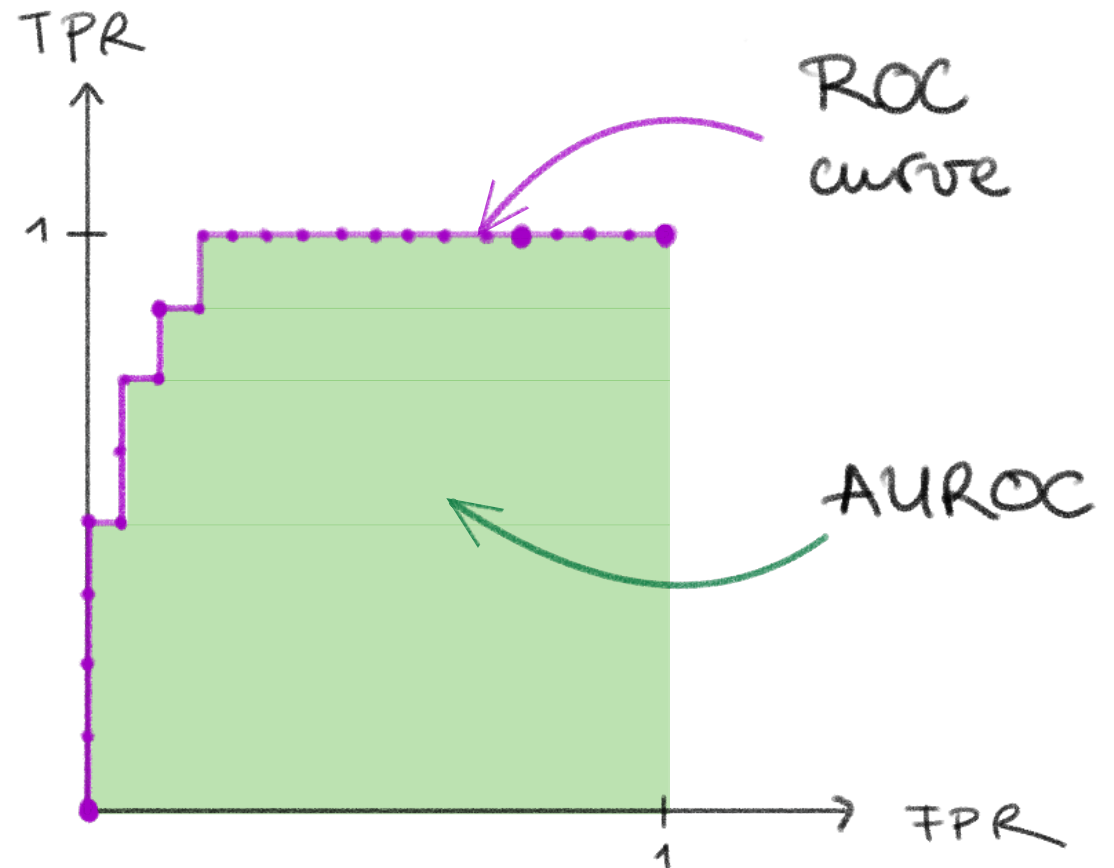


# EVALUATION METRICS



$$FPR(z) = 0$$

$$TPR(z) = 0$$



# Quiz

1. What would a "perfect" score look like?  
And the ROC curve/AUROC associated?
2. Same question for a "random" score.
3. Same question for a "bad" score.
4. Can we generalize the ROC curve/AUROC to the case of probability distributions?

# SOLUTIONS

# BENCHMARKING



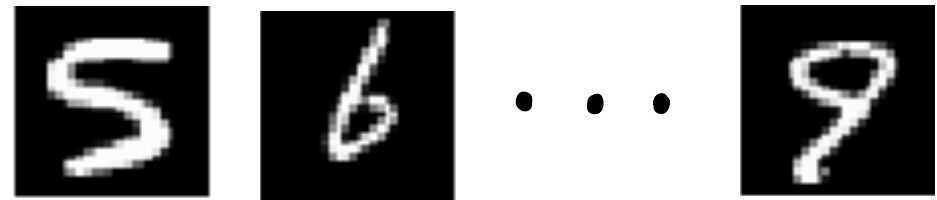
Lack of datasets for high-dimensional data (images, text)

see for example MVTecAD

Solution 1.



normal



anomalies

# BENCHMARKING



Lack of datasets for high-dimensional data (images, text)

see for example MVTEC AD

Solution 2 .

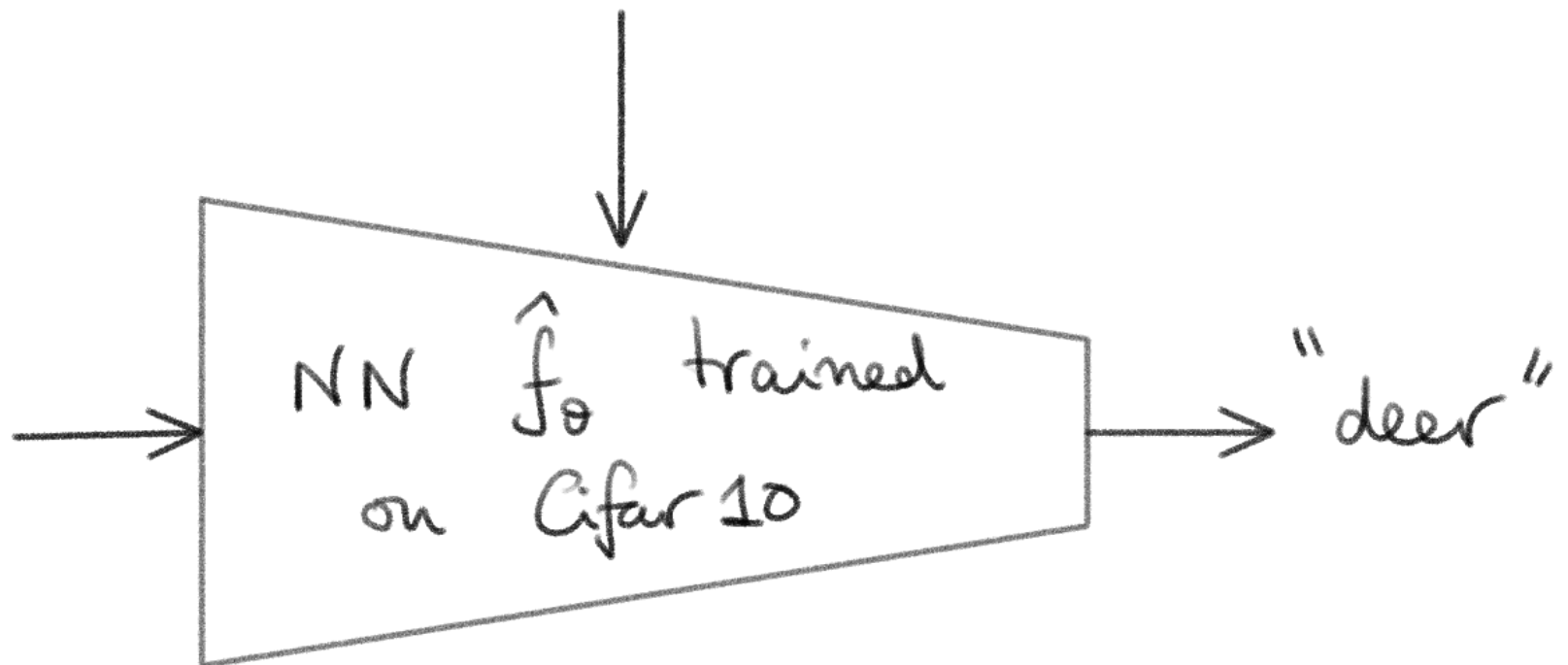
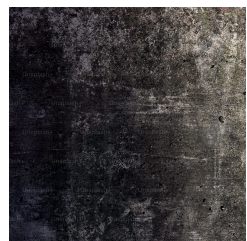


MNIST → normal



CIFAR10 → anomaly

# OUT-OF-DISTRIBUTION DETECTION



# OoD = AD ?

→ They accept similar mathematical formulations, but:

1. In OoD detection the space of anomalous images can be much larger
2. In OoD detection we have access to feature representations via  $\hat{f}_\theta$

# OoD SCORES

For a classifier  $\hat{f}_\theta$ :

$$\hat{f}_\theta : \mathcal{X} \xrightarrow{\hat{g}_\theta} \mathbb{R}^D \xrightarrow{\hat{h}_\theta} \mathbb{R}^d$$

feature space  
(usually the penultimate layer)

logit space

→ logit-based scores:  $s(x) = \tilde{S}(\hat{f}_\theta(x))$

→ feature-based scores:  $s(x) = \tilde{S}(\hat{g}_\theta(x))$

# LOGIT-BASED SCORES

→ Maximum Logit Score MLS

$$\tilde{S}(z) = -\max_{1 \leq c \leq C} z_c$$

→ Maximum Softmax Probability

$$\tilde{S}(z) = -\max_{1 \leq c \leq C} P_c(z)$$

# LOGIT-BASED SCORES

→ Energy

$$\tilde{S}(z) = -T \log \sum_{c=1}^d e^{\tilde{\theta}_c / T}$$

tunable  
hyperparameter

→ Entropy

$$\tilde{S}(z) = - \sum_{c=1}^d P_c(z) \log P_c(z)$$

# FEATURE-BASED SCORES

→ Deep K-nearest neighbor (DKNN)

Given  $z_1, \dots, z_N$  and  $z_*$  :

↑  
fitting dataset

↑  
test point

(i) Compute  $d_1 = d(z_1, z_*), \dots, d_N = d(z_N, z_*)$

(ii) Rank  $d_{(1)} \leq d_{(2)} \leq \dots \leq d_{(N)}$

(iii) DKNN score  $\tilde{S}(z_*) = d_{(k)}$

hyper-parameter

# FEATURE-BASED SCORES

→ Mahalanobis

Given  $z_1, \dots, z_N$  and  $z^*$  :

↑  
fitting dataset

↑  
test point

(i) Fit  $Q$  Gaussians:  $\mathcal{N}(\mu_c, \Sigma)$ ,  $c=1, \dots, Q$

$$\hat{\mu}_c := \frac{1}{N_c} \sum_{i: y_i=c}^N z_i, \quad \hat{\Sigma} = \frac{1}{N} \sum_{c=1}^Q \sum_{i: y_i=c} (z_i - \hat{\mu}_c)(z_i - \hat{\mu}_c)^T$$

# FEATURE-BASED SCORES

→ Mahalanobis

(ii) Mahalanobis score:

$$\tilde{S}(z) = \max_{1 \leq c \leq C} (z - \hat{\mu}_c)^T \hat{\Sigma}^{-1} (z - \hat{\mu}_c)$$

Exercise - If  $C=1$ ,  $\hat{\mu}_c = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$  and  $\hat{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ ,

draw  $A \subseteq \mathbb{R}^2$  given by  $A = \{z : \tilde{S}(z) \leq 1\}$

# RESOURCES

## Surveys

1. A unifying review of deep and shallow anomaly detection,  
Ruff et al. 2021



2. Generalized out-of-distribution detection, Yang et al. 2021



# RESOURCES

## GitHub repositories

1. OpenOOD



2. ADBench



# CHALLENGES

- Representation learning for AD
- Characterization of anomalies
- Interpretability of AD
- Creation of a unified methodology for AD research:  
benchmarks, metrics,...

# Quiz

1. Why is AD important for ML/DL?
2. Name 2 classes of AD algorithms.
3. Write down the formula for 1 metric.
4. Write down 1 sentence stating 1 thing you understood well.
5. Write down 1 question with something you would like to know more about.